Devon Traffic Insights

A FastHTML + Plotly reference dashboard for origin–destination, speed, and journey-time analytics across Devon, Torbay and Plymouth.

What this product delivers

This application demonstrates software that offers origin and destination traffic data, speed and journey time data for Devon county (including Torbay and Plymouth) from GPS signal data — or an equivalent source such as mobile-network probes.

Out of the box it ships with a deterministic synthetic generator so the system can be demonstrated, tested and benchmarked end-to-end without any vendor contracts. Each data module is designed as a drop-in adapter so the synthetic feed can be replaced by a Vodafone Analytics / O2 Motion / HERE / TomTom / Streetlight extract with no code changes above the data layer.

Demo KPIs (synthetic 14-day window)

Zones tracked	18 Devon towns + Torbay + Plymouth
Origin–destination pairs	306
Total trips / day	~8,200
Monitored road corridors	12 (M5, A38, A30, A380, A386 …)
Journey-time observations	~23,000
Synthetic GPS pings	5,000

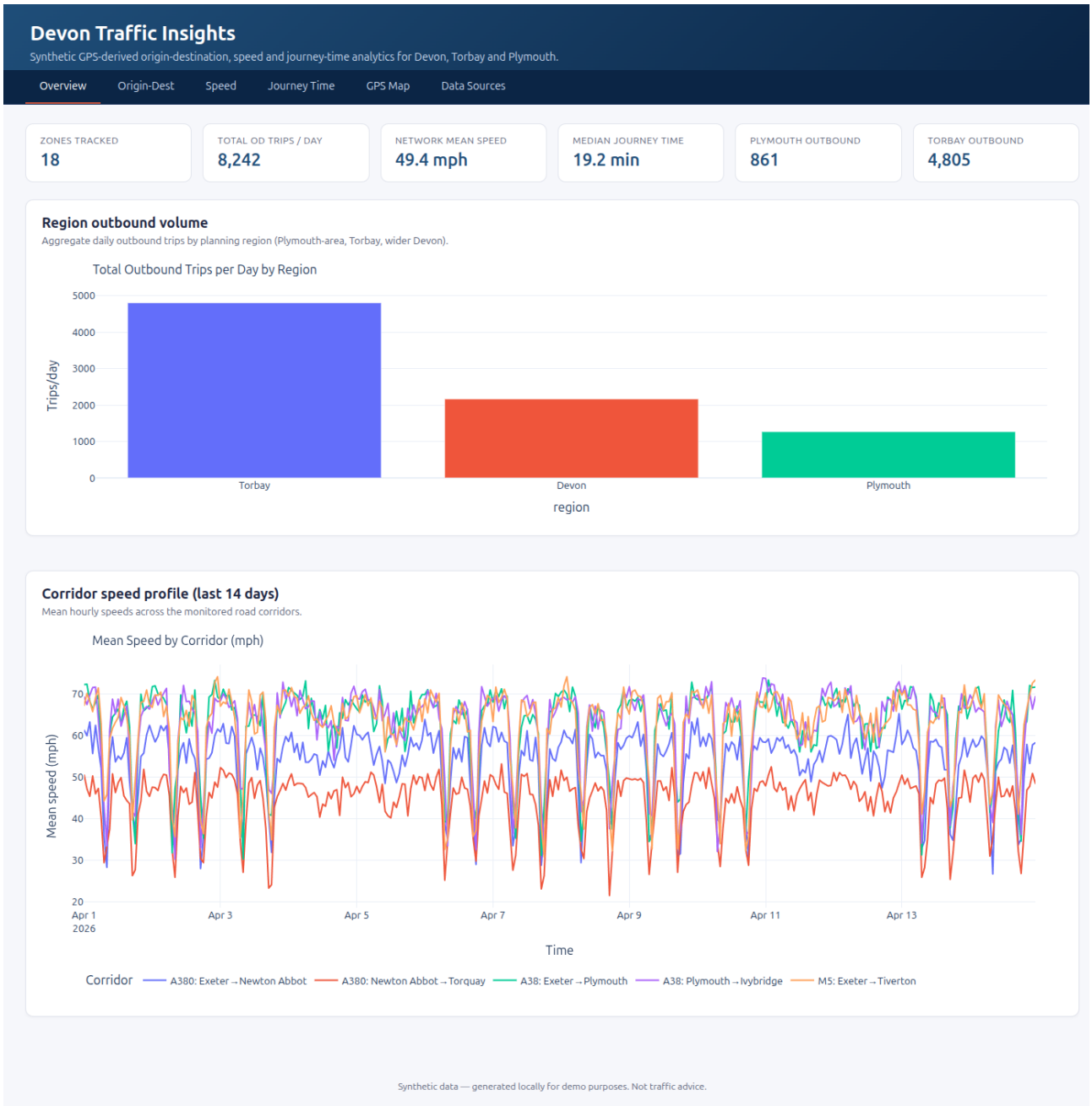
How this guide is organised

The following pages walk through each of the six dashboard sections exactly as a new user would see them — the screenshots are captured from the running FastHTML app via Playwright.

Overview

Executive summary of the Devon network.

Six KPI tiles summarise the demo dataset — zones tracked, total OD trips per day, network-average speed, median journey time, and outbound volumes for the Plymouth and Torbay travel-to-work areas. Two anchor charts follow: a regional outbound-volume bar, and a 14-day speed time-series across the key corridors (A38, A380 and M5) that makes the commuter-peak pattern immediately visible.

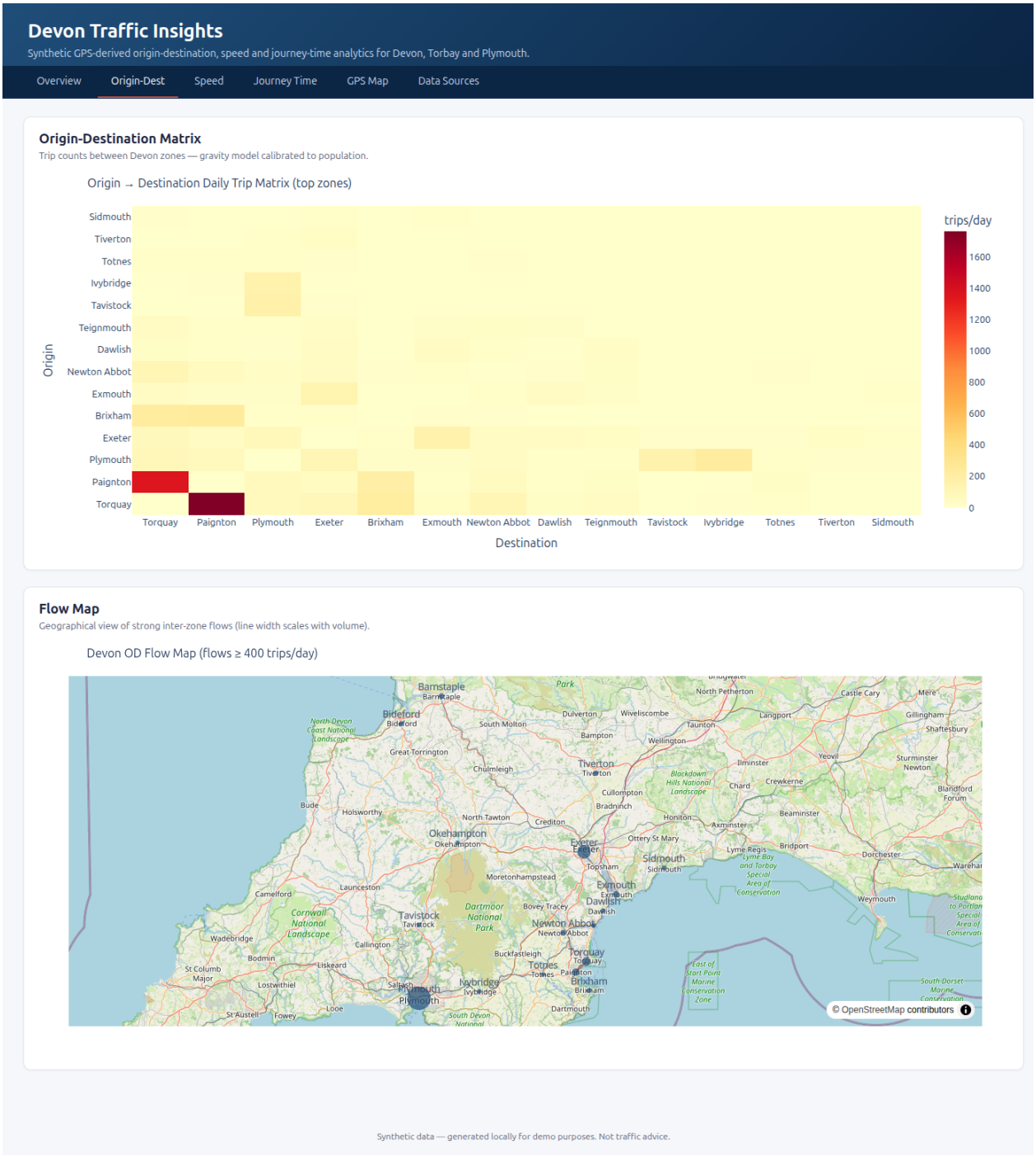


Screenshot: 01-overview.png

Origin–Destination

Where the trips go.

The heatmap shows daily trip counts between the 14 highest-volume Devon zones. Strong intra-Torbay flows (Torquay↔Paignton↔Brixham) and the Plymouth↔Ivybridge corridor are clearly visible. Below, the interactive flow map overlays every OD pair above 400 trips/day onto OpenStreetMap, with line width scaled to volume — making regional structure obvious at a glance.



Screenshot: 02-od.png

Speed

How fast the network actually runs.

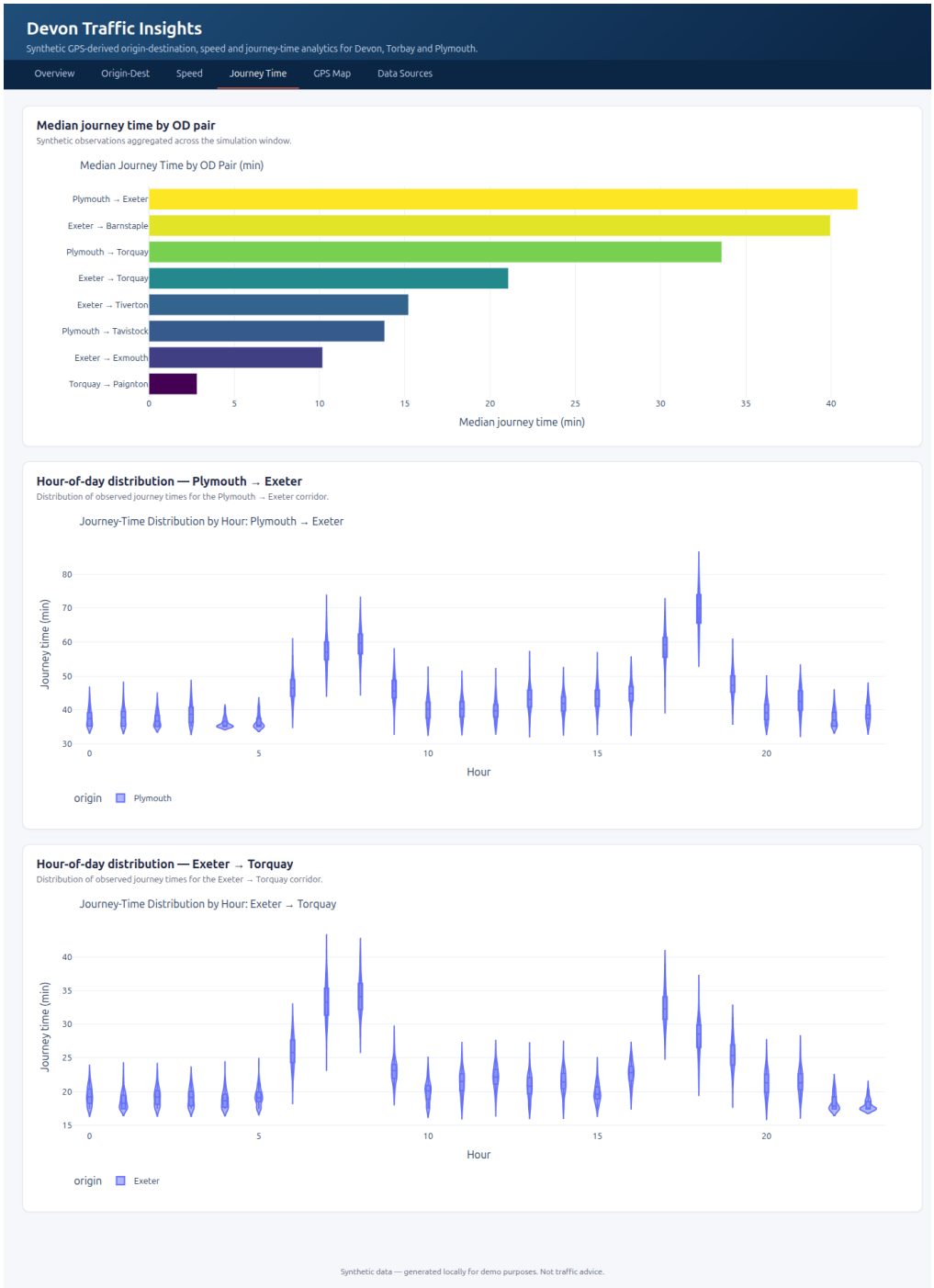
Top chart: hourly mean speed per corridor over the simulated 14-day window — weekday rush-hour troughs are clearly separated from weekend patterns. Bottom chart: the same data collapsed onto hour-of-day, which is the view operations teams typically use to tune signal timings, set variable speed-limits or plan roadworks windows. A30 and A379 show the sharpest commuter dips.

Screenshot: 03-speed.png

Journey time

How long trips take, and how that varies by hour.

The bar chart ranks the eight tracked OD pairs by median journey time. Below, violin plots show the full hour-by-hour journey-time distribution for Plymouth → Exeter and Exeter → Torquay. The distribution width tells you reliability, not just the mean — the key operational question for any journey-time product.

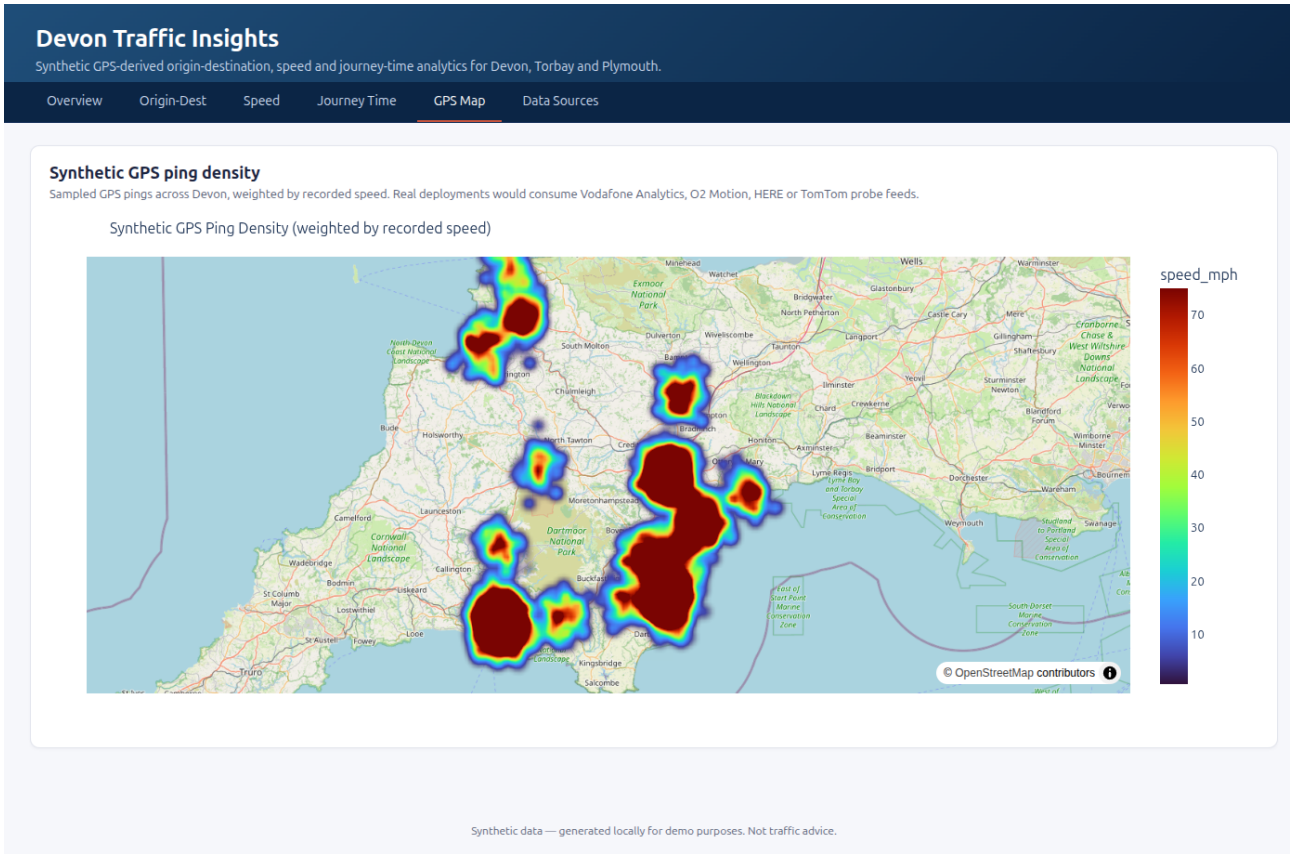


Screenshot: 04-journey.png

GPS map

The ping-level view that a real probe feed would populate.

A density heatmap of synthetic GPS pings across Devon, weighted by recorded speed. In production this layer is fed by a Vodafone Analytics / O2 Motion / HERE / TomTom probe extract, or by raw device-level feeds from Cuebiq / Veraset / Huq / Streetlight. The UX stays identical; only the data adapter changes.



Screenshot: 05-map.png

Data sources

Three build paths and the vendor shortlist.

The application ships with an in-app catalogue comparing (1) a solo build on DCC's own ANPR / Bluetooth / UTMC assets, (2) a solo open-data + ML-synthesis approach, and (3) a commercial probe-data integration (Vodafone, O2, HERE, TomTom, INRIX, Streetlight). A 4-step integration pattern — adapter → parquet cache → MSOA reconciliation → Census-calibrated blend — closes the page.

Devon Traffic Insights

Synthetic GPS-derived origin-destination, speed and journey-time analytics for Devon, Torbay and Plymouth.

Overview | Origin-Dest | Speed | Journey Time | GPS Map | Data Sources

Data source integrations

Swap the synthetic generator for any of the feeds below. Free/open sources need no contract; commercial feeds give finer granularity and population-calibrated expansion.

Build paths we compared

Path 1 — Solo, using DCC's own data

Depends on what Devon County Council will share. Typical DCC assets:

- **ANPR cameras** — Bluetooth/ANPR traces can build OD matrices for corridors they cover.
- **MOVA / SCOOT / UTMC** signal + loop data — link-level speed and flow (no OD).
- **Bluetooth beacons** (e.g. BlueTruth) on key corridors — journey-time series.
- **Existing traffic counts + DfT Road Traffic Statistics** — AADT by link.

Assemble OD from ANPR + Bluetooth via matrix-estimation (gravity / Furness / path-choice models) — a legitimate transport-modelling technique. **Gap:** rural Devon coverage is thin; no cross-county trip inference without GPS.

Path 2 — Solo, using open data + ML synthesis

DfT open data + OSM + Strava Metro (active travel only) + synthetic OD via ML (e.g. graph neural nets, gravity-model neural calibration, or entropy-maximisation against Census 2021 commute flows). Cheap and differentiated, but weaker on coverage score.

Path 3 — Commercial probe-data integration (recommended)

Best coverage and signal quality; requires contracts. See vendor list below.

Free / open

- **Department for Transport traffic counts** — AADT and hourly count data by road link. dft.gov.uk
- **National Highways WebTRIS** — motorway and strategic-road-network loop detectors (incl. M5). webtris
- **Devon County Council open data** — local road counts, bus patronage, cycle counters. devon.gov.uk
- **OS Open Roads / OS Open Zoomstack** — free road network geometry for routing.
- **TfL Unified API / BODS** — bus open data (NaPTAN, timetables, real-time). bus-data.dft.gov.uk
- **OpenStreetMap + Overpass / Valhalla** — free routing and turn-by-turn estimates.
- **ONS Census 2021 origin-destination** — commute flows by MSOA/LSOA (ground truth for OD calibration).
- **Strava Metro** — aggregated bike/pedestrian GPS (free for public bodies).

Commercial — probe / mobile / GPS

- **Vodafone Analytics** — UK mobile-network derived OD flows, dwell-times, visitor analytics.
- **Telefónica / O2 Motion** — aggregated O2 mobility insights, good UK coverage incl. South-West.
- **BT Active Intelligence / EE Mobility** — mobile probe data across EE's network.
- **HERE Traffic API & Probe Data** — real-time link speeds, historic 5-min bins, journey times.
- **TomTom Traffic Stats / O/D Analysis** — floating-car data with commercial-fleet GPS; strong on A-roads.
- **INRIX Roadway Analytics** — probe-vehicle speeds, incident feeds, OD analyses.
- **Google Maps Roads / Distance Matrix API** — live & typical travel times between coordinates.
- **Mapbox Movement** — anonymised mobile-SDK GPS aggregates.
- **Streetlight Data (Jacobs)** — mobile + nav-GPS OD matrices, widely used in UK DfT studies.
- **Cuebiq / Veraset / Huq Industries** — raw device-level mobile GPS panels (for custom OD builds).
- **Teralytics** — telco-derived trip chains and OD (global footprint).

Paid-ANPR / camera feeds

- **Devon & Cornwall Police ANPR** (via data-sharing agreement).
- **Vivacity Labs / VivaCity AI cameras** — classified multi-modal counts, journey times.
- **Clearview Intelligence M100** — Bluetooth/Wi-Fi travel-time network (commonly deployed on A38/A30).

Suggested integration pattern

1. Replace `devon_traffic.data.build_*` functions with adapters pulling from each feed's API or nightly dump.
2. Cache normalised parquet per feed under `data/`.
3. Reconcile zone geometries to a common MSOA or custom TAZ layer.
4. Blend: use Census 2021 OD as prior, calibrate with Vodafone/O2 expansion factors, validate with WebTRIS counts.

Synthetic data — generated locally for demo purposes. Not traffic advice.

Screenshot: 06-sources.png

Installing and running

Requirements: Python 3.10 or newer. All commands run from the repository root.

```
git clone <repo> traffic-data-analysis
cd traffic-data-analysis
python -m venv .venv
source .venv/bin/activate
pip install -e .
devon-traffic serve
```

CLI reference

```
devon-traffic --help
devon-traffic serve --port 5001
devon-traffic summary
devon-traffic generate-data --out data
devon-traffic export od --out od.csv
```

Regenerating this guide

Screenshots are captured via Playwright (see `tests/test_playwright.py`). The PDF is assembled by `scripts/make_pdf.py`, and the animated GIF by `scripts/make_gif.py`.

All figures in this guide are generated from synthetic data. Do not cite for operational decisions.